BEING A MONOMORPHISM IS FPQC LOCAL ON THE BASE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file descent.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted spans carry 1 navigation stub; the author's appeal to it is accompanied by the explicit reduction in the span.

Editorial note. The preamble span points at the chapter's own preamble.tex shipped with the frozen source, used verbatim as staged.

Editorial note. The canonical tag of this label does not occur among the tags of the published items of descent.tex, so it is not a published duplicate.

Allowed standard prerequisites (editorial).

- The coproduct of the base changed covering pieces maps onto the test scheme as an epimorphism. Same chapter, Lemma lemma-fpqc-universal-effective-epimorphisms, invoked at proof lines 6481-6483. Supplies the last step from equality after base change to equality before it.
- fpqc locality on the base and monomorphisms of schemes. Same chapter, the definition of fpqc localness and the notion of a monomorphism, named in the statement. Supplies the two objects the proof manipulates.

Source reference. Original lemma 0: sources/stacks-chow/tags.txt, source lines 1–1. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *The property $\mathcal{P}(f) =$"f is a monomorphism" is fpqc local on the base.*

Proof. Let $f : X \to S$ be a morphism of schemes. Let $\{S_i \to S\}$ be an fpqc covering, and assume each of the base changes $f_i : X_i \to S_i$ of f is a monomorphism. Let $a, b : T \to X$ be two morphisms such that $f \circ a = f \circ b$. We have to show that $a = b$. Since f_i is a monomorphism we see that $a_i = b_i$, where $a_i, b_i : S_i \times_S T \to X_i$ are the base changes. In particular the compositions $S_i \times_S T \to T \to X$ are equal. Since $\coprod S_i \times_S T \to T$ is an epimorphism (see e.g. Lemma 0) we conclude $a = b$. $\square$

Date: stacks-project:descent.tex:local-snapshot-20261012; GNU FDL 1.2 or later.